Femur Reconstruction with Probabilistic Shape Modelling

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***Abstract*—Probabilistic Shape Modelling(PSM) or Statistical Shape Modelling(SSM) is used in medical imaging where different models are generated for different parts of the body. These models are generated using mathematics, statistics, and machine learning. In this exercise, we have worked on the reconstruction of partial femur bones.**

***Index Terms*—Modelling,femur,statistical, probabilistic, reconstruction**

I. Introduction

The goal of this project is to generate a reconstructed femur model from the given partial femurs. The dataset contains 47 meshes and 47 landmarks for full femurs and 10 partial femurs that have to be reconstructed. Here we follow all the required steps that are Rigid Alignment, Building a GP model, Model fitting with Iterative Closest Points(ICP) building a PCA model out of the same. Then we proceed towards femur reconstruction.

II. Rigid Alignment

Downloading the data

We had downloaded the data for the femur project from the specified website and we registered and participated in the challenge. Post that, we received a download link to download the data. The data consisted of two folders named full-femurs and partial-femurs. Full femurs had 47 meshes ranging from 0 to 46 and for every mesh, there was a corresponding landmark in the landmark folder. The other folder named partial-femurs had 10 mesh files ranging from 47 to 56.

Rigid Alignment

For Rigid Alignment, we loaded the reference mesh(Fig. 1) and reference landmark to which the meshes and landmarks have to be aligned.

After performing the rigid alignment following result was obtained as all the femurs were aligned to the reference femur as expected. We set the color of the aligned femurs to green. Fig 4 represents all the aligned femurs after cleaning up the rigid femurs.

III. Building a GP model

In this exercise, we have built a Gaussian Process Model which could be used for establishing correspondence for further exercises. In our shape model, we have used a scalar

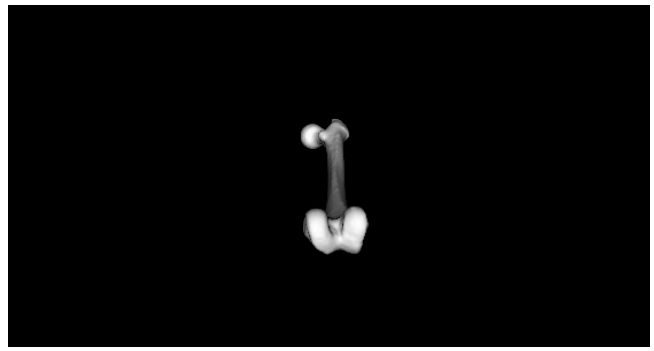

Fig. 1. Reference mesh that was loaded

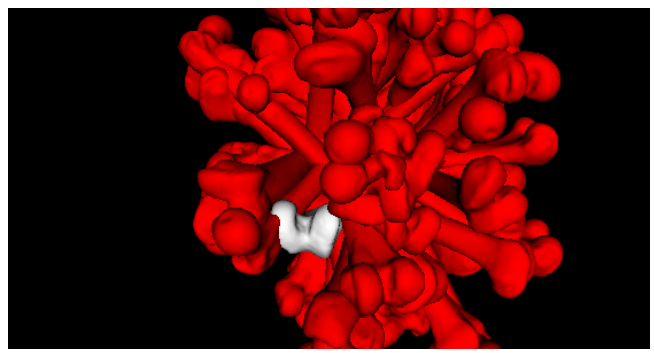

Fig. 2. 47 red rigid femurs and 1 reference femur in white

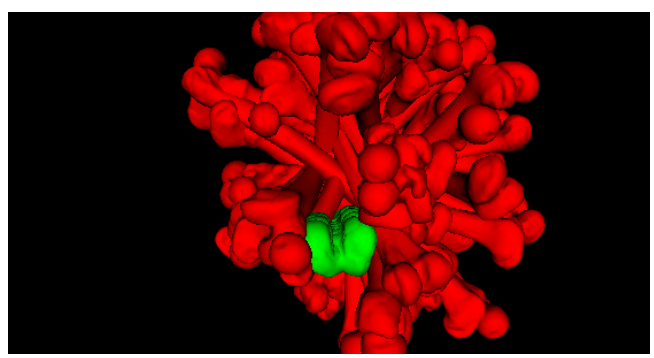

Fig. 3. 47 rigid femurs with 47 aligned femurs

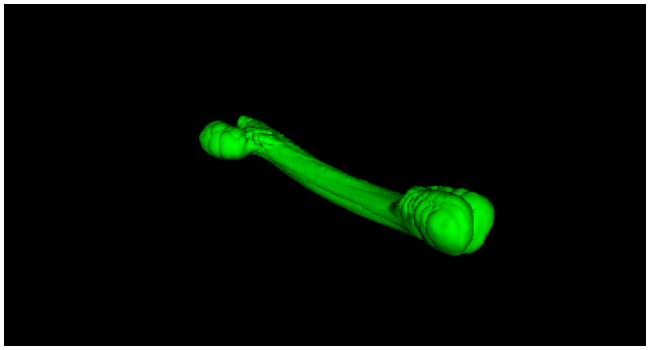

Fig. 4. 47 aligned femurs

kernel to build a matrix and to define a diagonal kernel. Let's define a scalar valued Gaussian Kernal k_s: $R^x X R^y \rightarrow R$

$$k_s(x, x') := sexp(-\frac{||x - x'||^2}{\sigma^2}) \tag{1}$$

$$k(x, x') = \begin{bmatrix} k_s(x, x') & 0 & 0 \\ 0 & k_s(x, x') & 0 \\ 0 & 0 & k_s(x, x') \end{bmatrix} \tag{2}$$

In Fig 5. we have the mean GP model that was generated for different values of σ and s. Fig. 6 and 7 display sample

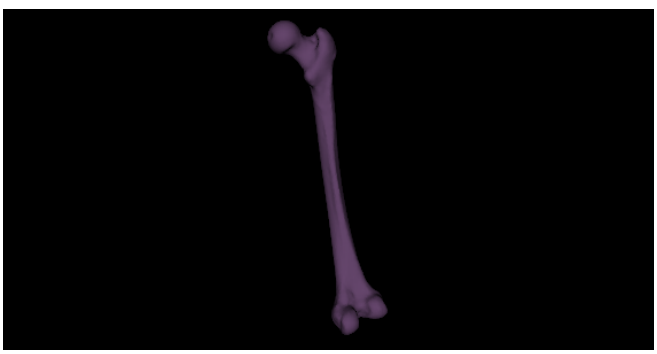

Fig. 5. Mean GP model

deformations with different scalar valued parameters on the diagonal kernel. Following are the kernel combinations we

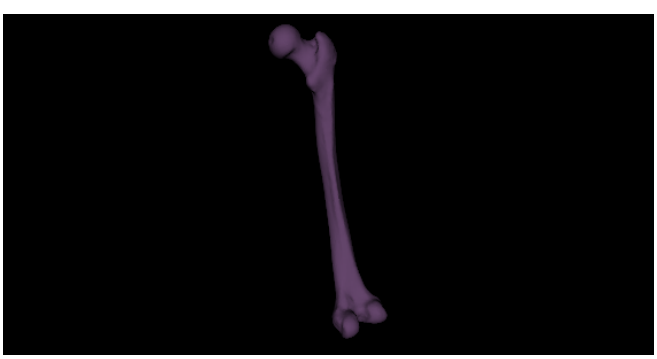

Fig. 6. GP model sampled with deformations

experimented with

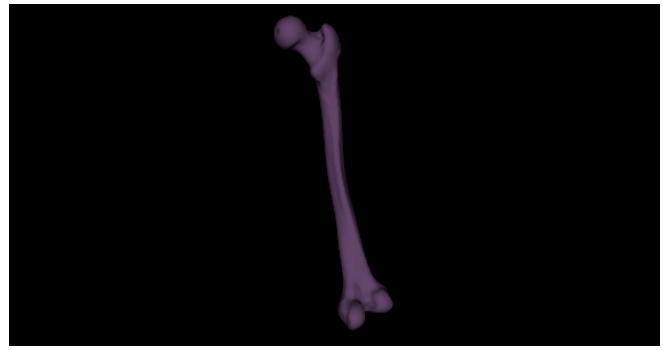

Fig. 7. GP model sampled with deformations

- We added 3 scalar valued kernels and used them in a diagonal kernel.
- We also performed experiment with addition of two diagonal kernels.
- We also performed experiment using scale and smoothness.
- We also performed multiple deformation changes with different kernel combinations like we removed the y and z plane completely i.e when kernel for y and z plane is set to 0. For example:

$$k(x, x') = \begin{bmatrix} k_s(x, x') & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \tag{3}$$

Fig 8. shows that this removes the deformations on the y and z direction and we have a smoother result.

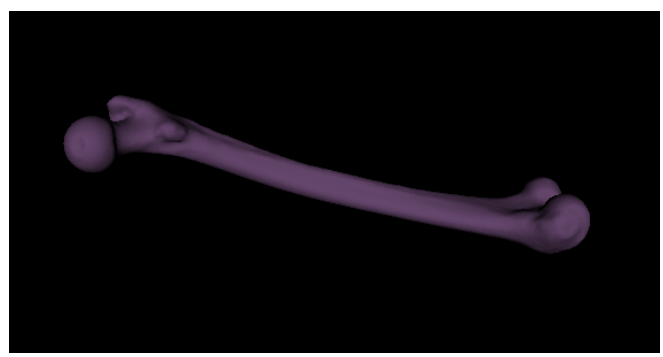

Fig. 8. After removal of deformations in y and z directions without deformation.

Following observations can be made based on the questions in the exercise:

- The chosen reference shape is a good reference shape because we are able to generate plausible deformations of the femur for the random values in the Gaussian Process.
- For fitting a model to a complete femur, the model should represent not only valid but also implausible femur shapes.
- We also observed that if we set GP mean to zero and set variance to zero, we will have a model with zero

deformations. Hence, higher the covariance, higher the deformations.

- We also combined scale and smoothness. When we increase the scale we observed larger deformations.[1]
- We can also have other combinations of kernels as well.

IV. Establishing correspondence and build a PCA model

In this exercise, we used the Gaussian Process model that was generated in the last exercise to establish correspondence to the sample femurs. After establishing the correspondence we build a statistical shape model of the femur.

- We worked through the usage of non-rigid ICP to establish correspondence to develop a registration algorithm. We had used the aligned meshes and landmarks that were generated from the rigid alignment step. ICP is generated where it is assumed that corresponding points are always the closest points in a mesh.[1]
- Here we create a posterior model for all the aligned meshes based on regression data and these posterior models are saved.
- We visualized the data in wireframe representation for all the meshes. Fig. 9 is sample registered meshes with wireframe representation.

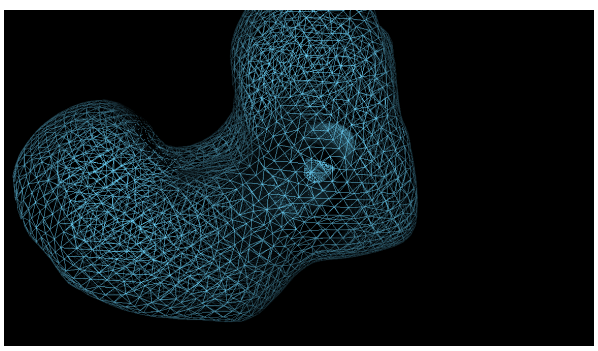

Fig. 9. Registered mesh wireframe representation

- Fig. 10 is an overlap of the registered mesh and the target mesh, where the red-colored mesh is the registered mesh and white is the target mesh. Fig. 11, 12, and 13 is the representation in the x, y, and z dimensions respectively.

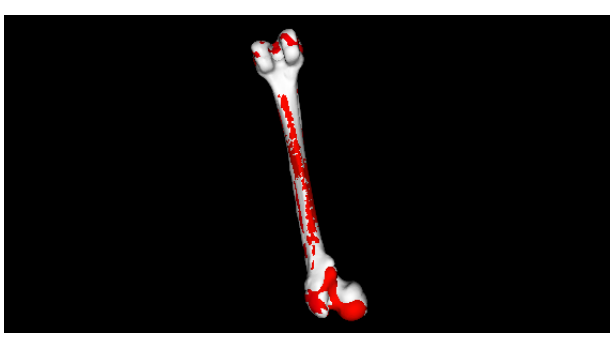

Fig. 10. A fitted mesh representation

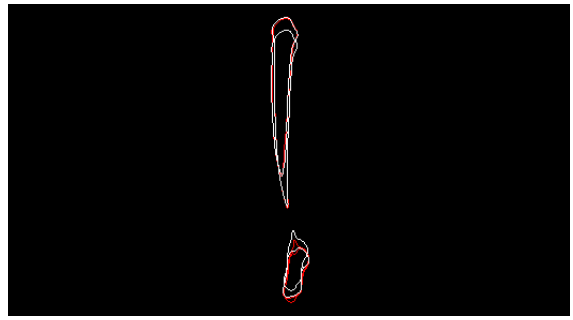

Fig. 11. Fitted mesh with x-Slice representation

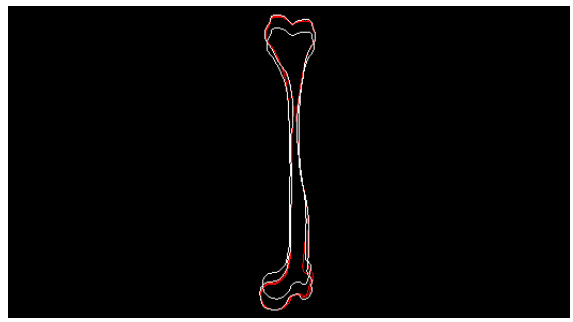

Fig. 12. Fitted mesh with y-Sclice representation

- In Fig 12. we can observe that mesh can fit in a better way as there are a lot of aberrations if we view from the y-dimension.
- We followed tutorial 6 of Scalismo.org and we tried generating the model from the data where we visualized the model and generated random samples.
  - Random samples that were generated did look like anatomically plausible shapes. Some samples were more deformed than expected. Fig 14,15 represents anatomically plausible samples whereas Fig.16 is a plausible sample but had much more visible deformation.
  - Fig.17 and 18 and represents situation when all sliders are pulled completely to the left and right respectively.
  - After pulling the all sliders to the left and right, the

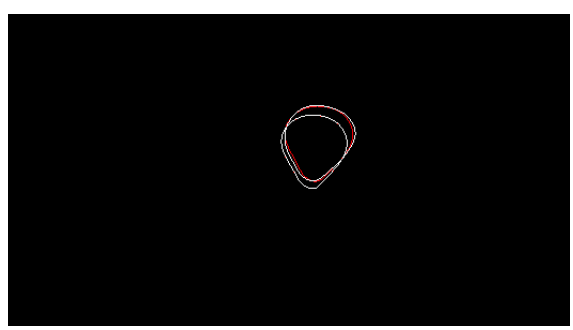

Fig. 13. Fitted mesh with z-Slice representation

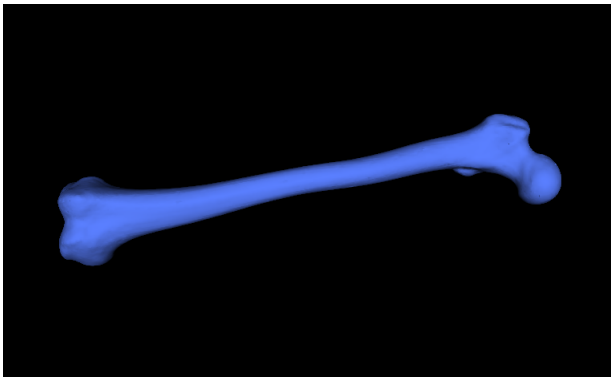

Fig. 14. First Random Sample with a anatomically plausible shape

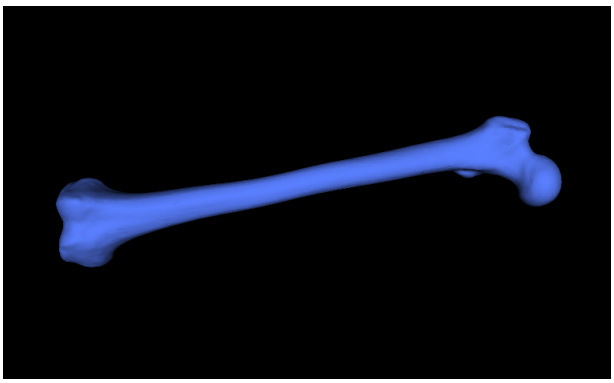

Fig. 15. Random SRandom Sample with a anatomically plausible shape

shapes are not anatomically plausible as you can see the fig. 16 and 17.

– We are not likely to obtain such a shape by random sampling where all values move towards one direction.

V. Femur Reconstruction

Using all the methods as mentioned from Exercises 1 to 3, we proceed to reconstruct partial femurs which is the whole objective of this project. Fig 18 is one of the partial femurs and Fig. 19 shows the corresponding reconstructed femur.

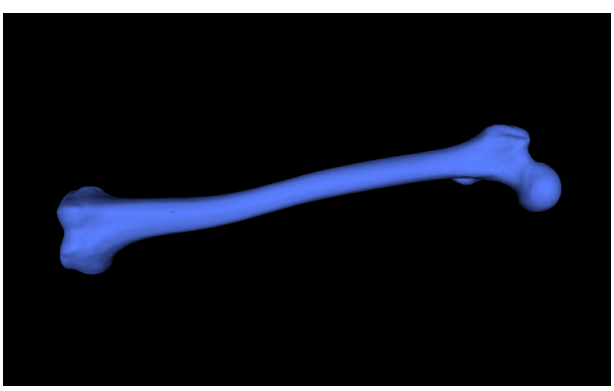

Fig. 16. Third Random Sample with a plausible shape but with a bend on the bone

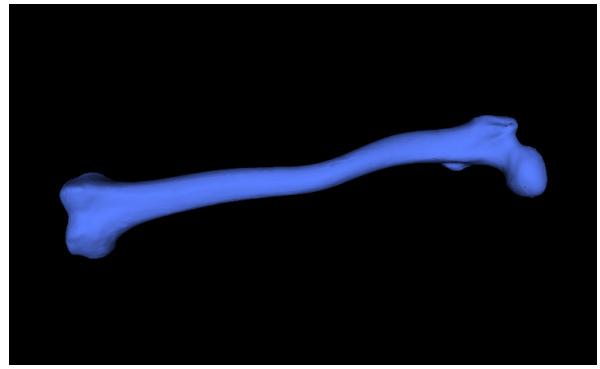

Fig. 17. Sample when all the sliders are pulled totally to the left

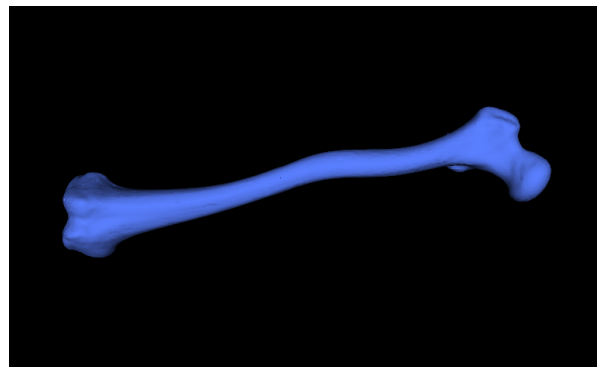

Fig. 18. Sample when all the sliders are pulled totally to the right

Following steps were followed for reconstructing the partial femur.

- We first loaded our PCA model.
- For every mesh we identified correspondences by attributing closest point on the target mesh.
- We define reference and target points.
- We then transform the moving mesh.
- Then we performed the non-rigid ICP registration with 500 iterations
- The above mentioned step was repeated again for all the partial femurs.
- Post completion of all these steps, we have all the partial-femurs reconstructed.
- As we can see in the Fig. 20, Femur 47 was best fitted. The Y-slice is also closely aligned
- Femur 56 was the worst fitted as represented in the Fig. 22.
- The STL files for all the femurs from 47 to 56 are enclosed with the report in the zip file.
- The femurs that end up having a bad means that the PCA model was not good enough for the femurs to be reconstructed to. If we have a better PCA model, we will have better results.

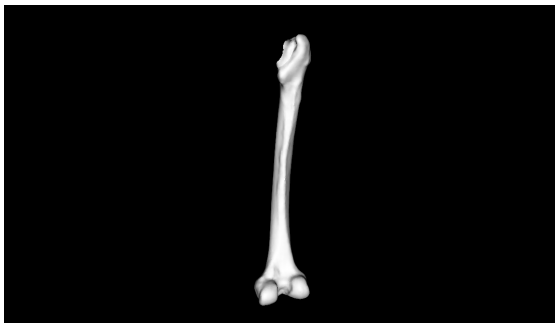

Fig. 19. Partial Femur

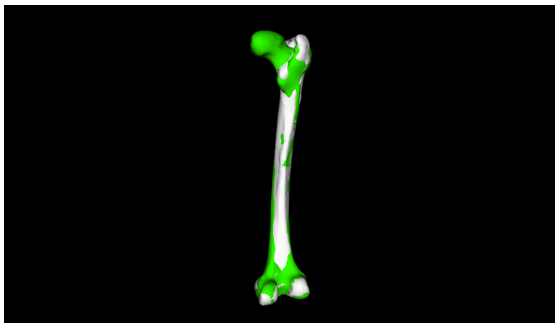

Fig. 20. Femur 47: Best final reconstructed femur

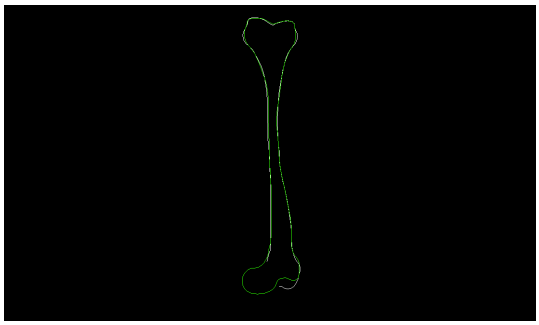

Fig. 21. Femur 47: Best Y-Slice of the reconstructed Femur

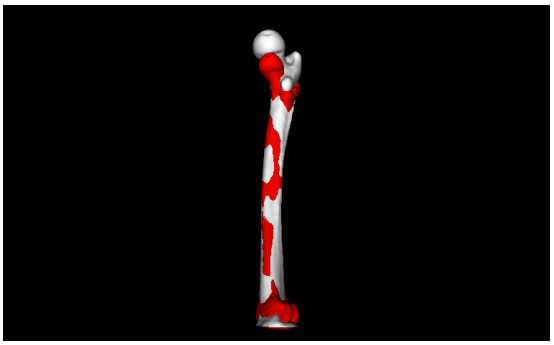

Fig. 22. Femur 56: Worst final reconstructed femur

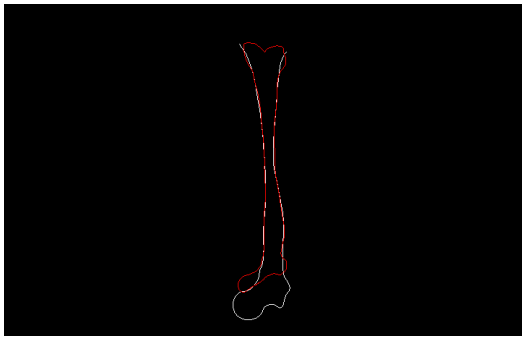

Fig. 23. Femur 56:Y-slice of worst final reconstructed femur

VI. RESULTS

Here we are visually representing the major step by step results:

- Fig 24. Set of 47 aligned femurs.
- Fig 25. Mean GP model.
- Fig 26. PCA model.
- Fig 27. Fitted Mesh using ICP algorithm
- Fig 28-37.Reconstructed Femurs 47-56

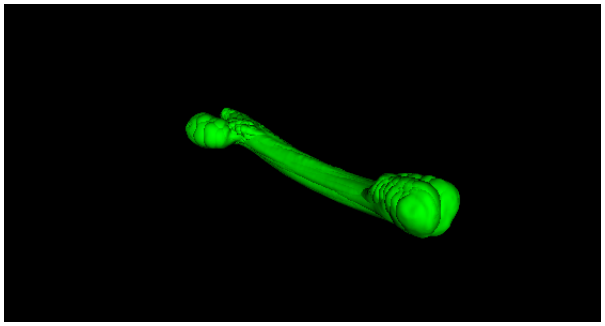

Fig. 24. 47 aligned femurs

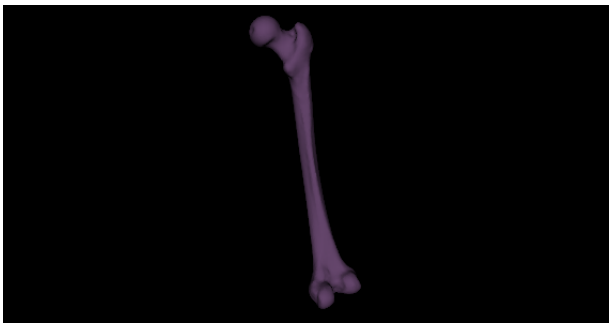

Fig. 25. Mean GP model

VII. DISCUSSION

Here we can conclude with following observations:

- We were successfully able to perform the rigid alignment.
- We also created a good GP model with the combination of different kernels for different values of scale and variance.

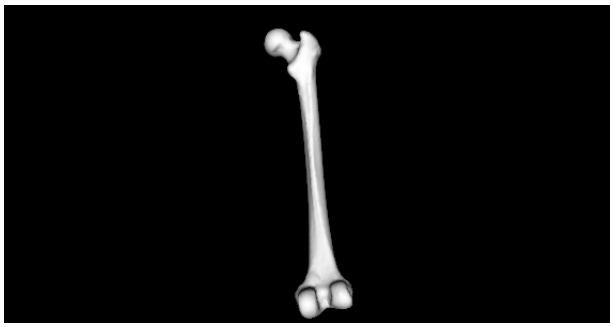

Fig. 26. PCA

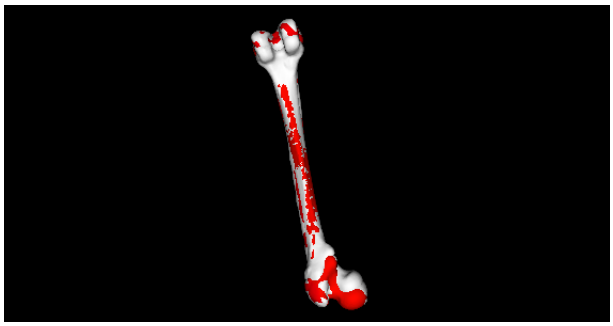

Fig. 27. A fitted mesh

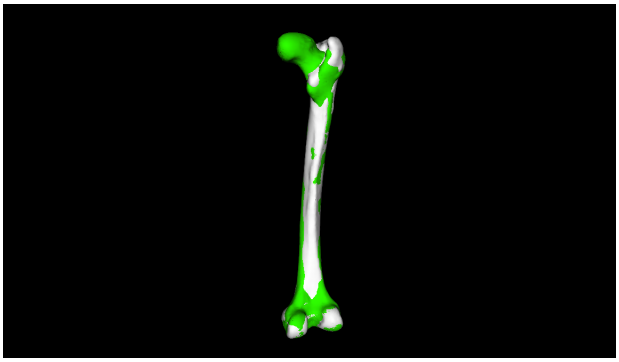

Fig. 28. Reconstructed Femur No. 47

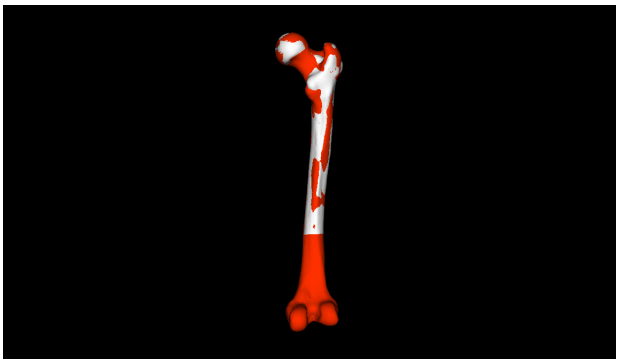

Fig. 29. Reconstructed Femur No. 48

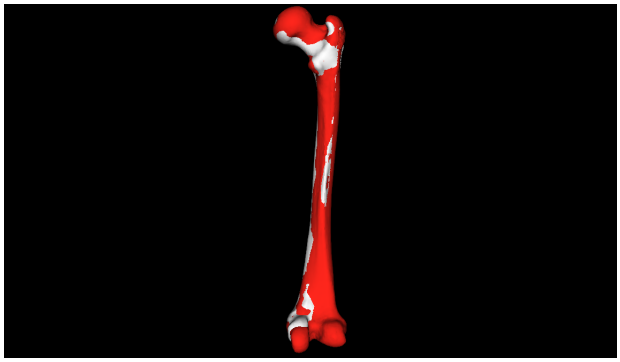

Fig. 30. Reconstructed Femur No. 49

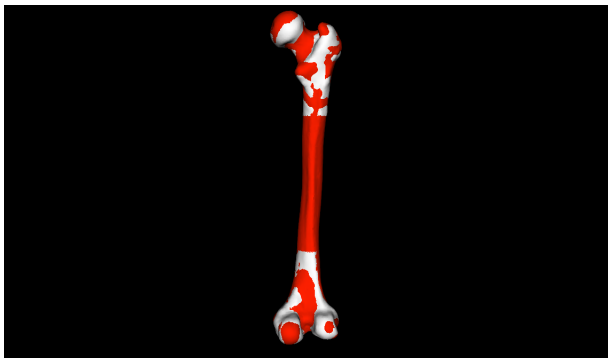

Fig. 31. Reconstructed Femur No. 50

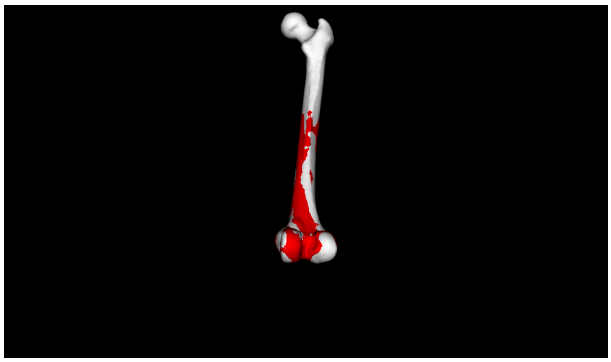

Fig. 32. Reconstructed Femur No. 51

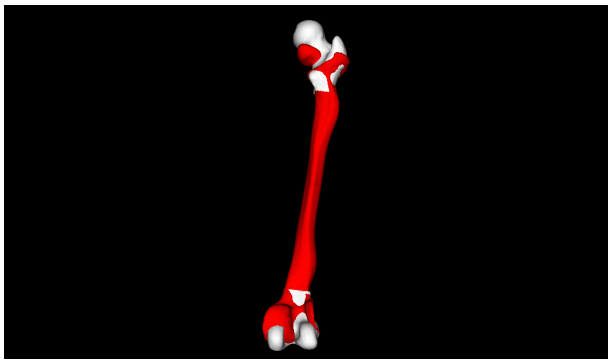

Fig. 33. Reconstructed Femur No. 52

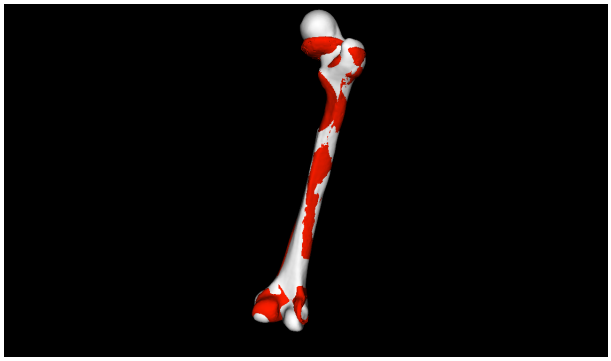

Fig. 34. Reconstructed Femur No. 53

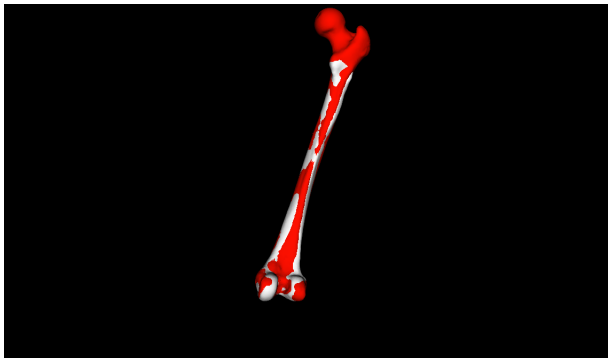

Fig. 35. Reconstructed Femur No. 54

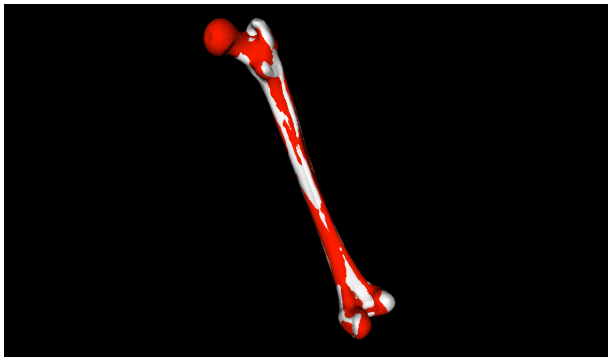

Fig. 36. Reconstructed Femur No. 55

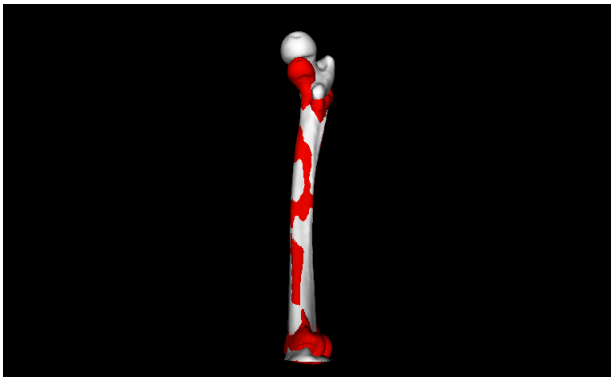

Fig. 37. Reconstructed Femur No. 56

- Non Rigid ICP algorithm also worked very well which did not result in any problems during ICP registration
- We were successfully able to reconstruct a full femur from a partial femur from the given data.
- The reconstructed full femur could be better as we can see in Fig.21, it has not been able to completely cover the femur. This means that we could have tried to prepare a better GP model.
- We can try to improve our PCA model, which could have been better and could have avoided some fitting issues as we can see that all partial femurs are not fitting perfectly like in Fig.22 for femur 56.

References

[1] Scalismo: https://scalismo.org/